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Review

Clonal analysis for understanding fate biases in developing embryos

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Multipotent progenitors that appear phenotypically similar often differ in the range of cell types they generate, which is typically described as fate bias. Traditional bulk lineage-tracing approaches, such as Cre/Lox systems, established qualitative progenitor–progeny relationships but offered limited insight into mechanisms governing tissue composition. Recent advances in lineage tracing combined with single-cell transcriptomics enable comprehensive characterization of clonal diversity at the whole-embryo scale, promising to provide mechanistic insights into developmental robustness. This review summarizes the current state of the art experimental and computational approaches in the field, with emphasis on emerging spatial and perturbation-based strategies in clonal biology.

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During development, cells undergo progressive differentiation, a process classically illustrated by Waddington's epigenetic landscape. This metaphorical topography maps the state space of potential cellular identities, where 'valleys' (stable attractors) and 'ridges' (unstable barriers) govern state transitions [1] (Figure 1a). Cells move across this landscape driven by sequential shifts in gene expression, where complex positive and negative feedback loops between different gene regulatory networks (GRNs) sculpt specific differentiation trajectories [2]. One of the central questions in developmental biology remains the molecular logic underlying cell fate choice — the bifurcations in the landscape where a cell must 'decide' between divergent paths. Identification and interpretation of this logic is fundamental to our understanding of development and is a prerequisite for the rational engineering of tissues and organoids [3].

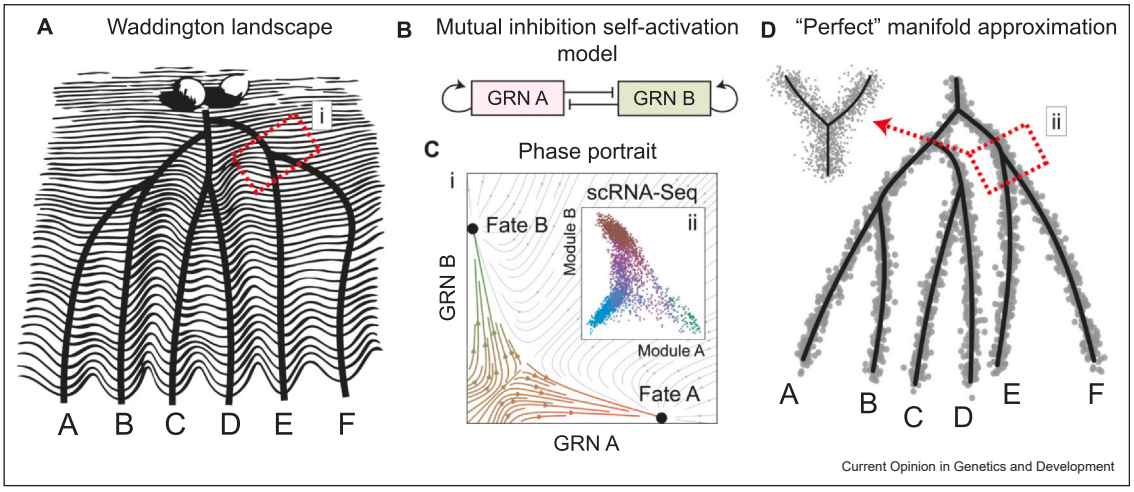
Several mechanisms have been proposed to explain lineage selection [5,6]. One prominent model describes fate decisions as a bifurcation resulting from competition between two antagonistic GRNs [7] (Figure 1b). This can be distilled into a three-step mechanism [8]:

1. Priming: Multipotent progenitors enter a state where competing GRNs for alternative fates are co-expressed at low or fluctuating levels.
2. Biasing: External signals or stochastic fluctuations tilt this equilibrium, rendering one GRN slightly dominant and shifting downstream fate probabilities without full commitment.
3. Lock-in (Commitment): Once a critical threshold is crossed, positive feedback and mutual inhibition solidify the choice, irreversibly committing the cell to a specific lineage.

This mechanism is a direct application of formal dynamical systems theory in developmental biology [9] (Figure 1c(i)) and has been experimentally validated across diverse biological contexts [7,10].

The advent of single-cell RNA sequencing (scRNA-Seq) has enabled the simultaneous measurement of tens of thousands of individual transcriptomes, allowing us to reconstruct molecular portraits of fate decisions at unprecedented resolution. The distribution of cells in high-dimensional gene expression space provides a robust approximation of the Waddington landscape [11] (Figure 1d). Notably, the transcriptional dynamics observed prior to bifurcation align closely with the mutual inhibition/self-activation model predictions [8] (Figure 1c (ii), from [4]). A critical implication of this model is that lineage biasing can occur long before overt differentiation markers appear; seemingly homogeneous progenitor cells may harbor distinct biases based on the subtle stoichiometric balances of competing GRNs [12,13]. This observation motivated the development of computational frameworks like FateID [14] (Figure 2a), which predicts fate biases by analyzing the activity of competing gene signatures in progenitor populations.

Figure 1



From Waddington's landscape to scRNA-Seq manifold. **(a)** The Waddington landscape (from [1] with modifications) showing average trajectories of differentiation into six cell types. The bifurcation (i) is highlighted in red. **(b)** Mutual inhibition/self-activation model with two competing GRNs. **(c)** (i) Phase portrait corresponding to the model in **(b)**, and (ii) a scatter plot (from [4]) showing the activity of two competing gene modules per cell in scRNA-Seq data undergoing a bifurcation. **(d)** 'Perfect' manifold approximation of the Waddington landscape from scRNA-Seq data. The bifurcation (ii) is highlighted in red.

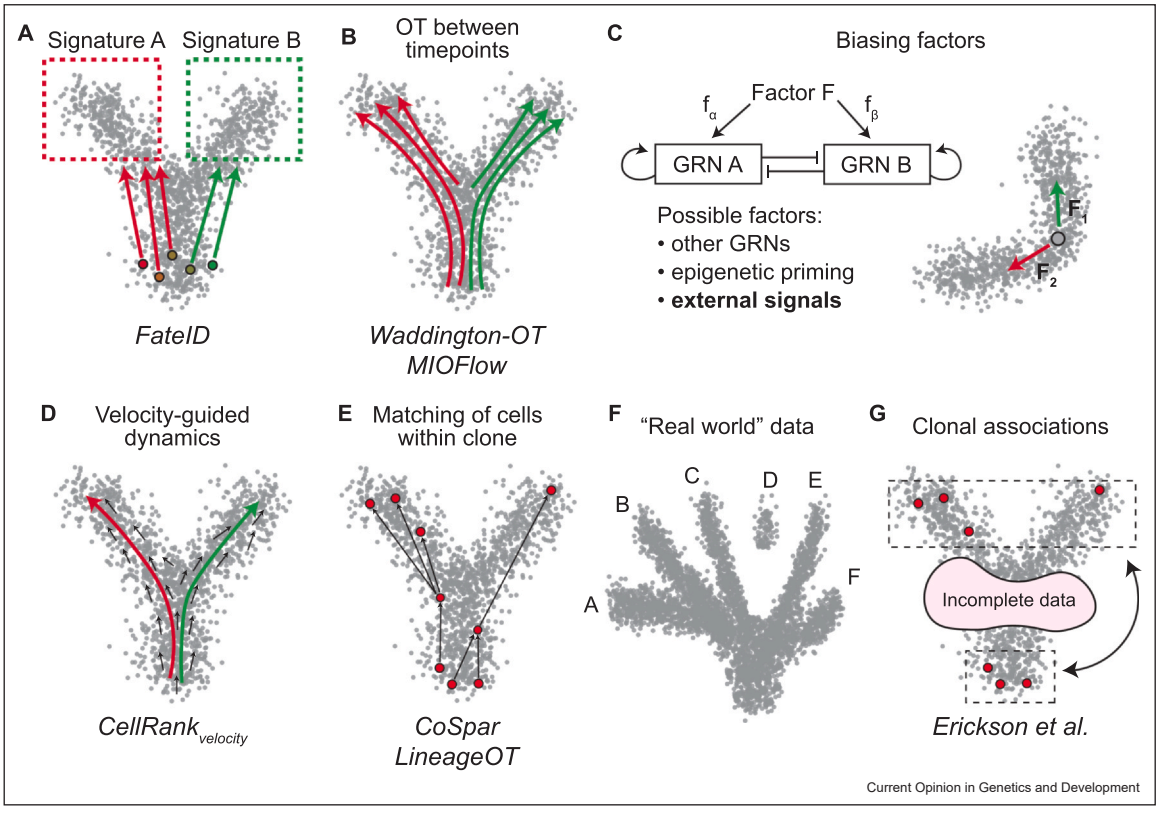
While signature-based approaches are intuitive, the nonlinear dynamics of GRNs often limit their predictive power. Consequently, methods such as MIOFlow [15] and Waddington-OT [16] utilize the geometry of the scRNA-Seq manifold itself to estimate fate biases, often employing optimal transport (OT) to model transition dynamics (Figure 2b). However, to resolve the directionality of movement, these methods typically require external temporal 'guidance', such as experimental timepoints or prior knowledge of progenitor–progeny relationships. The resulting models are thus highly sensitive to the underlying assumptions regarding manifold transitions [17].

Beyond these methodological sensitivities, differentiation is also orchestrated by extrinsic factors — including signaling contexts and spatial niche constraints [18] — and epigenetic priming [19] (Figure 2c). This can lead to 'hidden variable' scenarios where cells with identical transcriptomic profiles exhibit divergent differentiation dynamics [17], a phenomenon invisible to purely transcriptomic manifold methods. RNA velocity [20] emerged as a promising tool to infer directionality by analyzing the ratio of unspliced to spliced mRNA transcripts. In theory, this vector reflects the net effect of internal and external forces to predict the future state of individual cells. Methods like CellRank incorporate RNA velocity to model movement across the manifold, significantly enhancing fate prediction accuracy [21] (Figure 2d). Nevertheless, RNA velocity is subject to technical limitations and biological assumptions that can limit its reliability in complex systems [22,23]. Importantly, CellRank and related frameworks project velocity vectors onto a k-nearest-neighbor graph to compute cell-to-cell transition probabilities, meaning that the manifold topology directly shapes the resulting fate predictions even when velocity itself is accurate.

The limitations of purely transcriptomic manifolds have motivated the coupling of scRNA-Seq with lineage tracing [24]. While tracing does not capture continuous state transitions, it provides the essential 'ground truth' of cellular history in the form of the final fate distribution of a given cell's progeny. Lineage recording systems generally fall into two categories: evolving barcodes (e.g. CARLIN [25], DARLIN [26], scGESTALT [27], LINNAEUS [28], ScarTrace [29]), which utilize Clustered Regularly Interspaced Short Palindromic Repeat-induced scars to reconstruct deep phylogenies, and static barcodes (e.g. CellTagging [30], LARRY [13], TREX [31]), which label progenitor pools with unique, heritable sequences. For the purpose of fate-mapping, static barcodes are often preferred, as evolving clones can be computationally collapsed into static 'clades' for analysis.

In vitro systems further enable 'sister cell' assays, where a clone is split: one 'arm' reveals the progenitor's transcriptomic state while the other reveals its eventual fate. Computational frameworks like CoSpar (graph-regularized optimization) [32] and LineageOT [33] bridge these modalities to clarify progenitor dynamics based on progeny distributions (Figure 2e). Interestingly, despite the inclusion of clonal data, these methods often exhibit predictive accuracies similar to velocity-guided models

Figure 2



Approaches for identifying biasing correlates in progenitors. **(a)** Signature-based methods (FateID) score expression signatures associated with alternative fates in progenitor cells. **(b)** Manifold-based OT methods (Waddington-OT, MIOFlow) find optimal matchings between cells at earlier and later timepoints. **(c)** Hidden factors — including other GRNs, epigenetic priming, and external signals — can produce divergent biasing outcomes from identical transcriptomic profiles (right), limiting purely manifold-based predictions. **(d)** Velocity-guided methods (CellRank) incorporate RNA velocity vectors to improve fate bias prediction. **(e)** Lineage-informed methods (CoSpar, LineageOT) use clonal relationships to link progenitor states to fate outcomes. **(f)** In practice, the manifold assumption is often violated in real datasets. **(g)** Manifold-free approaches directly test associations between progenitor gene expression and clonal fate distributions (Erickson et al.).

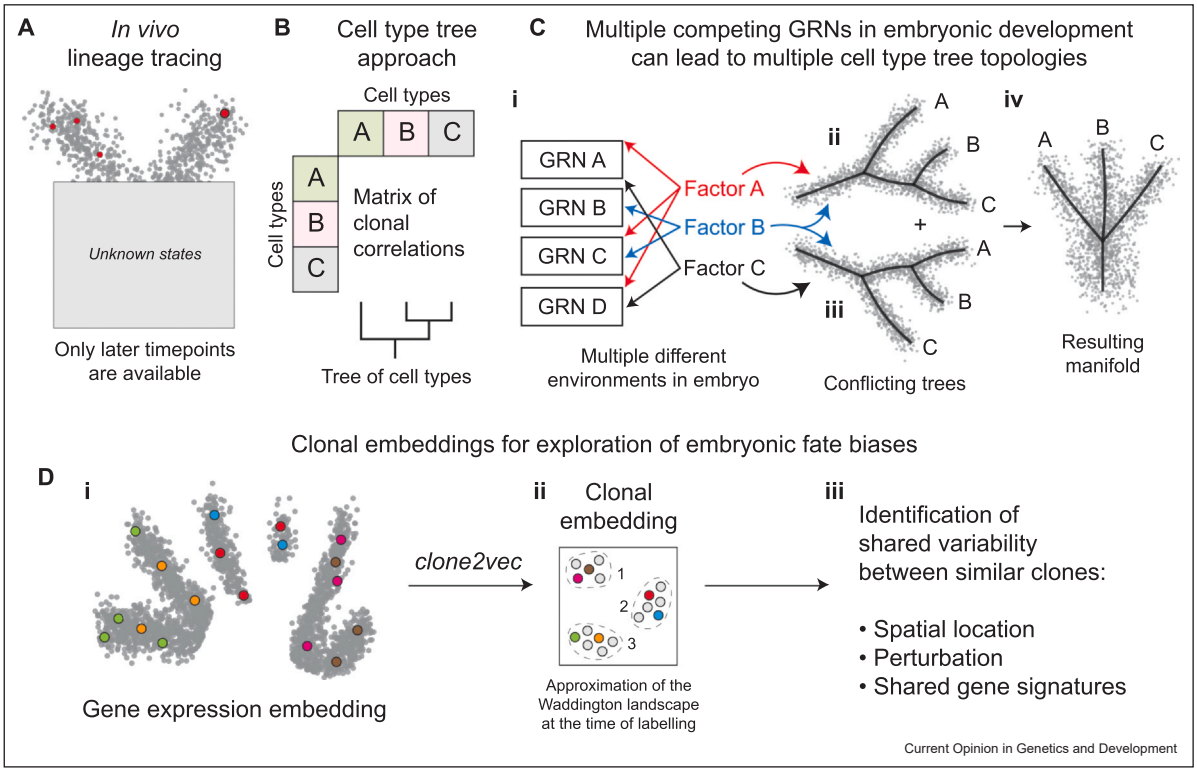
[34], likely reflecting inherent bottlenecks in the transcriptomic manifold approximation itself.

More broadly, however, the manifold is often an incomplete representation regardless of the analytical strategy. Measurement noise blurs the topography, leaving barrier heights and attractor depths poorly defined. Datasets are frequently sparsely sampled, missing critical intermediate states and violating the assumption of trajectory continuity (Figure 2f). The manifold assumption can be bypassed by directly testing associations between gene expression in the progenitor arm of a clone and the fate outcome of its differentiated arm (Figure 2g), but this comes with its own disadvantages — primarily the small number of clonally labeled cells in most datasets [13,32].

In the developing embryo, multi-timepoint clonal analysis remains technically elusive. With few exceptions where progenitors and progeny coexist — such as hematopoiesis [35] or mesenchymal differentiation [36] — direct application of described approaches is often impossible (Figure 3a). As a result, many *in vivo* studies are restricted to reconstructing cell type differentiation trees based on the covariation of clonal abundances across cell types [37,38] and similar approaches [39,40] (Figure 3b) or must rely on computational inference from the non-barcoded data [41,42]. Furthermore, *in vivo* barcoding is significantly more complex than *in vitro* applications — often barcode injection and manipulation are performed *ex vivo* before re-implantation, which limits its utility for studying undisturbed embryonic development.

The described 'differentiation tree' approach assumes a fixed, universal structure [43], an assumption that may not hold *in vivo*. Differentiation is heavily regulated by spatial signaling contexts, meaning the topology of the differentiation tree can vary significantly across different embryonic locations (Figure 3c). To address this, the computational method clone2vec groups clones by their

Figure 3



In vivo lineage tracing and tree-free approaches to fate bias identification. **(a)** *In vivo* lineage tracing typically captures only later timepoints, with progenitor states unobserved. **(b)** The cell type tree approach infers differentiation hierarchies from a matrix of clonal co-occurrence across cell types. **(c)** In embryonic development, multiple competing GRNs are modulated by distinct spatially varying factors (i), which can produce conflicting cell type tree topologies in different embryonic regions (ii, iii). The resulting scRNA-Seq manifold (iv) averages over these contexts, obscuring location-specific branching structures. **(d)** The clone2vec strategy bypasses tree reconstruction entirely. Clones observed at a later timepoint (i) are embedded by their fate distributions (ii). Clusters of clones with similar fate profiles are then analyzed for shared features that may underlie the observed biasing patterns (iii).

fate distributions to identify shared molecular features across clones with similar behaviors [36]. This 'tree-free' method does not require the presence of progenitor cells in the dataset, relying instead on the assumption that shared features across distinct clones reveal underlying regulatory mechanisms (Figure 3d).

clone2vec is not a direct competitor for methods that aim to identify biasing programs in the progenitor population; indeed, some of those methods may have higher sensitivity for that specific task. Rather, clone2vec operates at a different level of abstraction. Clonal embedding — the output of the algorithm — represents each clone as a point in a space of fate probabilities under a multinomial assumption. It treats each clone as a random variable sampled from the 'true' clonal distribution in gene expression space, shifting the question from "what biases a particular clone toward one or another fate?" to the more general "which biasing patterns do we observe across the dataset?" — all without relying on discrete cell type assignments. The resulting clonal clusters can reveal stereotypical fate distributions encompassing whole ensembles of cell types, which better reflects the reality of embryonic development, where borders between cell types are often fuzzy, while at the same time fine spatial regulation produces consistent proportions of cell types across different body regions. Variation of clonal patterns can uncover smooth or abrupt transitions in lineage composition across space, thereby revealing how local anatomical structures and boundaries emerge and providing insight into the fundamental principles of morphogenesis.

In line with this logic, a critical shared feature between clones with similar behavior — and a primary readout of signaling context — is spatial location [36]. Because most clones (excluding highly migratory populations like neural crest or immune cells) exhibit restricted movement [44], sister cells share a history of molecular and physical cues. Linking fate distribution to spatial coordinates represents a conceptual shift: rather than asking if a clone is biased, we ask if the tissue contains 'bias fields' — spatially patterned regimes that amplify,

suppress, or redirect intrinsic lineage tendencies. Recent advances in imaging-compatible RNA barcoding are now bridging the gap between high-resolution spatial maps and clonal identity [31,45].

Ultimately, both lineage tracing and RNA velocity provide correlates of fate biases rather than causes. Moving from descriptive to causal models requires the interpretation of GRN activity and its modulation by molecular signals and 'master' transcription factors. Several computational methods have been developed to predict cellular responses to perturbations as a proxy for fate transitions. Benchmarking studies suggest that CellOracle [46], a regression-based approach, currently outperforms other methods in this task [47]. However, the most definitive way to identify the drivers of fate bias is through systematic perturbations. Although analyzing tissue composition following a perturbation provides a direct readout of its influence on fate decisions, an additional clonal layer of information can give insights into the level of heterogeneity in mechanisms behind the bulk compositional effect, opening doors to further explanation of the observed effects [36].

The combination of spatial transcriptomics, lineage tracing, and mosaic perturbation screening (particularly for ligand-receptor interactions) represents the next frontier in developmental biology. Together, these modalities can map the bias fields introduced above — the exact biasing patterns triggered by specific signals within their native spatial contexts — closing the loop between correlative fate prediction and mechanistic understanding.

CRediT authorship contribution statement

Sergey Isaev: Conceptualization, Visualization, Writing – original draft, Writing – review & editing. **Igor Adameyko:** Conceptualization, Visualization, Writing – original draft, Writing – review & editing, Project administration, Supervision, Funding acquisition.

Data Availability

No data were used for the research described in the article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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